

Lesson 4-3, Period 1 (Observation Day) – Teacher Plan

Multiply and Divide Rational Expressions • Friday 2026-05-15 • Period F (65 min)

Date	2026-05-15 (Fri)	Grade Level	11–12
Subject	Algebra 2 / Period F	Section	Topic 4 (Rational Functions) – Lesson 4-3, Period 1
Common Core State Standards	HSA-APR.D.6 (rewrite simple rational expressions in different forms); HSA-APR.D.7+ (understand that rational expressions form a system closed under operations analogous to rational numbers); HSA-SSE.A.2 (use the structure of an expression to identify ways to rewrite it).		
Topic Goals	Use the structure of rational expressions to rewrite them in different forms while tracking domain restrictions. Build and interpret rational expressions that arise from real-world ratios (geometry, probability, efficiency). Recognize that simplifying a rational expression does not change its domain restrictions.		
Essential Question	What stays the same and what changes when we rewrite a rational expression in simplified form?		
Materials	Student packet (4 pp.); highlighters (one per student); pencils; teacher iPad (for circulating + capturing work); whiteboard + projector; visible timer; reference: Savvas SE/TE for 4-3.		
Content Objective(s)	I can identify the values that must be excluded from the domain of a rational expression. I can simplify rational expressions by factoring and canceling common factors. I can build and interpret rational expressions that arise from real-world ratios.	Criteria for Success / Formative Assessment	On Lupe Carnival Part B, student writes the simplified expression $\frac{x}{6(x+1)}$ AND states $x \neq 0, -1$ with a physical interpretation. On Part C, student substitutes $x = 4$ correctly AND makes a defensible fairness judgment. Self-Reflection: “I can explain why the original denominator (not the simplified one) determines the domain.”
Language Objective	Students will explain why a domain restriction must come from the <i>original</i> denominator (not the simplified one), verbally and in writing, with the support of structured talk-time, models, and real-time scaffolding/questioning.	Criteria for Success / Formative Assessment	Use the printed sentence frames to articulate (a) the simplified expression and (b) the excluded values with a reason from the original expression. Use the comparison frame “rational expressions are similar to rational numbers because _____.” Self-Reflection: “I used at least one sentence frame today to explain my thinking.”
Modifications / Support for IEP students	Tasks pre-chunked (Part A / B / C scaffolds on packet). Sentence frames printed inline. Extended time embedded (groups of 2–3 work at own pace inside time block). Highlighters provided for the 3-Reads. Teacher checks in 1:1 during circulation lap 2.	Supports for ELL students	3-Reads routine (teacher reads first, then students read + highlight). Sentence frames on every long-explanation prompt. Visual TikZ diagrams for both modeling tasks. Word bank in Share/Summary: <i>domain restriction, simplified, equivalent, closed, undefined</i> . Group A/B/C structure builds peer support.

Lesson Part	Teachers will. . .	Students will. . .	Questions to consider. . .
Do Now (10 min) <i>Function-family match + open/closed circle review, DOK 1–2</i>	Hand out student packet at the door. Project Page 1 (Do Now). Part 1 (5 min): Read aloud the prompt “Match each graph A–D to its function family.” Set timer 3 min silent matching, 1 min pair check, 1 min cold-call to share answers. Anticipated: A=quadratic, B=exponential, C=linear, D=rational. Part 2 (4 min): Read the two number lines. Ask “what’s different?” Set 2 min silent + 2 min pair share. Surface: closed = included, open = excluded. Notation: Line 1 is $x \geq -3$; Line 2 is $x \neq 2$. Bridge to Explore & Reason: “Today’s family is <i>rational</i> – and the open circle on Line 2 shows up everywhere when we restrict domains.”	Look at each graph; recall what its parent function looks like. Match each to a word from the bank. Compare answers with elbow partner. Read both number lines; notice the open vs closed circle. Write a domain expression for each line.	<i>How can you engage students in the lesson?</i> <i>How can you activate prior knowledge of function families?</i> <i>How are all adults intentionally planned for?</i>
Explore & Reason (8 min) <i>$y = x + 2$ with $x = 2$ excluded – DOK 2 conceptual gateway</i>	Project the graph of $y = x + 2$ from packet Page 2. Read aloud Parts A, B, C once. Set timer 5 min silent independent work. Then 2 min pair share. Cold-call 1–2 pairs to share Part C answer. Bridge to Example 6: “When the denominator equals zero, the function is undefined. That’s a <i>domain restriction</i> – and we’ll see SA/V ratios in the next two problems.”	Read the prompt silently. Answer Part A (domain of $y = x + 2$). Sketch Part B (function with $x = 2$ excluded). Write Part C using the printed sentence frame. Volunteer (or be cold-called) to share.	<i>How are you eliciting prior reasoning before introducing new content?</i> <i>How will you handle students who don’t yet think of rational?</i>
Example 6 (10 min) <i>Cube vs Cylinder SA/V (guided) – DOK 2 procedural</i>	Project Page 3 (Example 6). We do (4 min): Walk through the cube side-by-side with students. Show $V = (2x)^3 = 8x^3$, $SA = 6(2x)^2 = 24x^2$, $SA/V = 3/x$. You do (5 min): Students compute the cylinder on their own using $V = \pi r^2 h$, $SA = 2\pi r h + 2\pi r^2$, with $r = x$, $h = 2x$. Circulate to verify they substitute correctly. Comparison (1 min): Cold-call. Both should be $3/x$ – the surprise. Surface: “equal ratios means choose by other criteria (cost, manufacturing).”	Take notes as teacher works the cube. Compute V , SA , and SA/V for the cylinder independently. Compare with the cube ratio. Use the sentence frame to articulate the design conclusion.	<i>What will you explicitly model and how?</i> <i>How will you check that students substitute $r = x$ and $h = 2x$ correctly?</i> <i>Are students ready for the harder Apply problem after this?</i>

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Apply (10 min) #35 Cylinders – Make Sense & Persevere, DOK 2 3-READS ROUTINE (full scaffold)	3-Reads (2 min total): • Read 1 (1 min): <i>Teacher reads aloud.</i> Students put pencils down and listen. Ask: “What is this problem about?” • Read 2 (1 min): Students re-read silently and <i>highlight</i> the formulas, the dimensions of A and B, and the question. Ask: “What information do we need?” Part a (5 min): students work <i>individually</i> on A’s V/SA. Teacher circulates with iPad noting interesting work. Gather responses from groups A, B, C in turn. Do not give the answer – collect thinking, surface differences. Part b (5 min): 2 min individual + 3 min turn-and-talk. Share out + review. Part c (3 min): TPS structure. Surface ratio comparison + design implication using the sentence frame on the page.	Listen during Read 1 (no writing). Highlight key info during Read 2. Work Part a individually for 5 minutes. Be ready to share when called by group letter (A / B / C). Work Part b individually for 2 minutes, then discuss with elbow partner for 3. Apply same TPS pattern to Part c. Use the sentence frame to share the comparison.	<i>What will you explicitly model and how?</i> <i>How will you know if students are ready to work alone?</i> <i>How will you elicit student thinking without revealing the answer?</i> <i>How are all adults intentionally planned for?</i>
Performance Task (18 min) #36 Lupe Carnival – Model with Mathematics, DOK-3 SPINE 3-READS (gradual release)	3-Reads – gradual release (2 min total): • Read 1 (1 min): <i>Students read silently first</i> (no teacher read aloud). Ask: “What’s the question?” • Read 2 (1 min): Pair share: “What information do we need to build the probability?” Group work on Parts A, B, C in groups of 2–3. Teacher circulates with iPad, capturing build strategies, errors, and domain interpretations. Mid-explore checkpoint (10 min mark): brief pause – “Is your simplified expression $\frac{x}{6(x+1)}$? Anyone get something else?” Don’t reveal whether correct – surface the discussion. End of phase: cold-call by group letter – Group A shares Part A reasoning; Group B shares Part B (simplify + domain); Group C shares Part C (fairness judgment). Rotate based on iPad notes.	Read silently first (Read 1). Discuss with group what the question is asking (Read 2). Build the rational expression as a group on Part A. Simplify and state domain restrictions in physical context (Part B). Substitute $x = 4$ and judge fairness (Part C). Use the printed sentence frames. Share out one part per group letter.	<i>What will students do for most of the session?</i> <i>What will they practice?</i> <i>What type of academic conversations will students engage in?</i> <i>How will you know which groups need re-direction?</i> <i>How will you press for the DOK-3 lift (build → simplify → judge) without revealing the path?</i>
Share / Summary (4 min) Synthesis around throughline	Synthesize the day’s throughline: <i>“Domain restriction is the soul of a rational expression – it survives even when factors cancel.”</i> Connect Do Now → Apply → PT explicitly – every problem returned to the same idea. Preview Topic 4 LEHS Q#3 and Q#5 (asymptotes from rational expressions – they come from these same domain restrictions).	Listen to the synthesis. One pair re-states the throughline in their own words (cold-call). Use the word bank if helpful: <i>domain restriction, simplified, equivalent, closed, undefined.</i>	<i>How will students express what they have learned?</i> <i>How does today’s lesson connect to upcoming Topic 4 assessment?</i>

Lesson Part	Teachers will. . .	Students will. . .	Questions to consider. . .
Exit Ticket (5 min)	Direct students to Page 4 (#14 Use Appropriate Tools). Set timer 5 minutes. Collect at the door as students leave. Tonight: scan exits for misconceptions about $\frac{-6x^2+21x}{3x}$ vs. $-2x + 7$ at $x = 0$ – the should-be-a-hole vs. defined-point distinction. If many students miss it, open Monday’s Do Now with a bridge re-prompt.	Read #14 prompt. Use the sentence frames if needed. Complete reflection prompts (“learned” / “liked”). Hand to teacher at the door.	<i>How will students show what they’ve learned based on the objective?</i> <i>Are the sentence frames being used?</i>